

Aiden Lee

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EDUCATION

Northwestern University

B.S. Computer Science, 3.93 / 4.00 GPA

Coursework: Distributed Systems, Networking, ML, Scalable Architectures, Operating Systems, Compiler Construction

Spring 2027

Evanston, IL

SKILLS

Languages: Python, C++, Go, TypeScript/JavaScript, Java, SQL, C

Technologies: Git, AWS, MCP, React, Docker, Kafka, PostgreSQL, Django, Flask, Redis, LangGraph, MySQL

EXPERIENCE

Pinterest

Software Engineer Intern

June 2026 – Sept. 2026

Remote

- Shipped Kafka High Disk agent (**3-5x/day**) on petabyte-scale clusters, automating broker-set analysis and mitigation steps to aid oncall employees via Slack
- Building MCP and Skills tooling for Logging alerts (CDC GTID diffing), replacing error-prone manual runbooks
- Designing internal OpenClaw integration where PagerDuty alerts auto-invoke agents that reply in the alert thread

360 Privacy

Software Engineer Intern

Sept. 2025 – Dec. 2025

Remote

- Developed a LangGraph state machine for automated parser generation with human-in-the-loop validation, reducing manual data transformation effort for unstructured datasets
- Designed and implemented a parallelized producer-consumer data pipeline, enabling memory-efficient streaming ingestion of **1GB+ files** into Pinecone with **5x throughput increase**
- Built an identity resolution system using union-find clustering and weighted phonetic similarity algorithms, achieving **98.4% precision** on duplicate detection on **86K+ records**

CodeHS

Software Engineer Intern

June 2025 – August 2025

Chicago, IL

- Built core utilities for Bool, enabling 1k+ users to generate websites from a prompt
- Engineered high-performance async job processor with Django + Redis for **99%+ reliable stream completion**
- Cut cloud costs by 50%** by eliminating duplicate startup instances spawning parallel workers and buckets
- Shipped promo engine powering all discount codes and feature flag system for PM-run A/B testing

PROJECTS

Compiler | C++23

Mar. 2026 – June 2026

- Built 5-stage optimizing compiler in **21K+ lines of C++**, lowering C-Like source through IRs to x86-64 assembly
- Won class compiler competition** at **536ms** runtime – **18x faster than GCC**, **2x faster** than previous winner
- Implemented 10+ SSA optimizations (SCCP, GVN, LICM, etc), advanced register alloc, and instruction tiling

NU Esports Discord Bot | Python, PostgreSQL, Docker

Sept. 2025 – Present

- Shipped commands giving **1500+ students** real-time game room visibility, replacing walk-in availability checks
- Built self-hosted GGLeap proxy reusing attendant session JWT to unlock PC state and reservation endpoints
- Built PostgreSQL reservation scheduler with conflict detection, room allocation, and prime-time quotas per team

NUFood | Go, TypeScript, React, PostgreSQL, AWS

Sept. 2024 – Sept. 2025

- Launched optimized dining hall app used by **500+ students**, featured in *The Daily Northwestern*
- Engineered a scalable backend API in Go, **reducing latency by 90%** compared to available alternatives
- Deployed on AWS with a batched data pipeline to cache all pages upfront for instant UI filtering

Content Creation | YouTube

Dec. 2024 – Present

- Create live programming content for **2,000+ subscribers**, generating **125,000+ views**